BACHELOR OF COMPUTER APPLICATION

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 13

Integration Part II

Table of Contents

SL.No	Topic		Fig No / Table / Graph	SAQ / Activity	Page No
1	Introduction		-	-	3 - 4
	1.1	Learning Objectives	-	-	
2	Integration by Parts		-	-	5 - 7
	2.1	Formula for Integration by Parts	-	-	
	2.2	Steps for Integration by Parts	-	-	
3	Integration of rational Function		-	-	8 - 11
	3.1	Understanding Rational Functions	-	-	
	3.2	Techniques for Integrating Rational Functions	-	-	
	3.3	Direct Integration	-	-	
	3.4	Polynomial Long Division	-	-	
	3.5	Partial Fraction Decomposition	-	-	
	3.6	Integrating Rational Functions with Repeated Linear Factors	-	-	
4	Working Examples		-	-	12 - 22
5	Summary		-	-	23 - 24
6	Self-Assessment Questions		-	1	25 - 26
7	Terminal Questions		-	-	27
8	Answers		-	-	28 - 29

1. INTRODUCTION

Welcome back! In Unit 12, we delved deeply into the initial concepts and techniques of integration. We explored how integration serves as the inverse process to differentiation, helping us accumulate quantities and understand areas under curves. You learned various integration techniques, starting with basic substitution methods which simplified integrals involving composite functions. We also tackled the integration of circular and inverse circular functions, where you mastered handling trigonometric integrands. This foundational knowledge paves the way for more complex integration techniques that await you in Unit 13.

Now that we have a solid grasp of basic integration concepts, Unit 13, titled "Integration Part II," focuses on advancing these skills further. This unit is crucial because it introduces Integration by Parts—a technique derived from the product rule for differentiation. This method is particularly useful for integrating products of functions, where one component is easily integrable, and the other is easily differentiable. Understanding and mastering this technique will equip you with the tools to tackle more complex integrals and enhance your problem-solving capabilities in calculus, essential for fields such as engineering, physics, and economics.

Approaching Unit 13, I encourage you to keep a curious and analytical mindset. Start by reviewing the formula for Integration by Parts and understanding its derivation, as this will be pivotal. Practice is key—engage with the step-by-step examples provided in the coursework to build your confidence. Try to identify the best choices for u and dv in practice problems, as this selection can simplify your work significantly. Utilize group discussions and interactive problem-solving sessions to deepen your understanding and resolve any doubts. This unit is not just about learning to perform integrations; it's about developing a deeper insight into the interplay between different calculus concepts that you've learned so far.

1.1. Learning Objectives

At the end of this unit, you will be able to:

- *Recall the formula for Integration by Parts.*
- *Explain the concept of Integration by Parts and its relationship with the product rule for differentiation.*
- *Perform integrations using the Integration by Parts technique on given functions.*
- *Differentiate between when to use basic substitution and when to use Integration by Parts based on the function provided.*

INSPIRED BY LIFE

2. INTEGRATION BY PARTS

Integration by parts is a powerful technique used to integrate products of functions. It is particularly useful when the integrand is a product of two functions where one function can be easily differentiated, and the other can be easily integrated. This method is derived from the product rule for differentiation.

2.1. Formula for Integration by Parts

The formula for integration by parts is:

$$\int u\,dv = uv - \int v\,du$$

Here, u and dv are parts of the original integrand udv. The goal is to choose u and dv such that du and v are simpler to work with.

2.2. Steps for Integration by Parts

1. Choose u and dv: Select parts of the integrand u and dv such that du and v are simpler.
2. Differentiate u: Find du by differentiating u.
3. Integrate dv: Find v by integrating dv.
4. Substitute into the formula: Use the integration by parts formula to rewrite the integral.
5. Simplify and integrate: Simplify the resulting integral and integrate if necessary.

Examples:

1. Integrate $\int xe^x\,dx$

   Solution:

   Choose u and dv;

   $$u = x, \qquad dv = e^x\,dx$$

   Differentiate u:

   $$du = dx$$

   Integrate dv:

   $$v = e^x$$

   Substitute into the formula:

   $$\int xe^x\,dx = xe^x - \int e^x\,dx$$

   Simplify and integrate:

   $$\int xe^x\,dx = xe^x - e^x + c$$

$$\int xe^x\,dx = e^x(x^{-1}) + c$$

2. Integrate $\int x\,ln(x)\,dx$

   Solution:

$$u = ln(x)$$

$$dv = x\,dx$$

$$du = \frac{1}{x}dx$$

$$v = \frac{x^2}{2}$$

$$\int x\,ln(x)\,dx = ln(x)\cdot\frac{x^2}{2} - \int \frac{x^2}{2}\cdot\frac{1}{x}dx$$

$$\int x\ln(x)\,dx = \frac{x^2\ln(x)}{2} - \int \frac{x}{2}dx$$

$$\int x\ln(x)\,dx = \frac{x^2\ln(x)}{2} - \frac{1}{2}\int x\,dx$$

$$\int x\ln(x)\,dx = \frac{x^2\ln(x)}{2} - \frac{1}{2}\cdot\frac{x^2}{2} + c$$

$$\int x\ln(x)\,dx = \frac{x^2\ln(x)}{2} - \frac{x^2}{4} + c$$

3. Integrate $\int e^x\,sin(x)\,dx$

   Solution:

   First Integration by Parts:

$$u = sin(x), \qquad dv = e^x\,dx$$

$$du = cos(x)\,dx$$

$$v = e^x$$

$$\int e^x\sin(x)\,dx = e^x\sin(x) - \int e^x\cos(x)\,dx$$

   Second Integration by Parts:

$$u = cos(x)$$

$$dv = e^x\,dx$$

$$du = -\sin(x)$$

$$v = e^x\,dx$$

$$\int e^x\,cos(x)\,dx = e^x\,cos(x) + \int e^x\,sin(x)\,dx$$

Putting it all together:

$$\int e^x \, sin(x) \, dx = e^x \, sin(x) - \left(e^x \, cos(x) + \int e^x \, sin(x) \, dx\right)$$

Solve for the original integral:

$$\int e^x \, sin(x) \, dx = e^x \, sin(x) - e^x \, cos(x) - \int e^x \, sin(x) \, dx$$

$$\int e^x \sin(x) \, dx + \int e^x \sin(x) \, dx = e^x \sin(x) - e^x \cos(x)$$

$$2\int e^x \sin(x) \, dx = e^x(\sin(x) - \cos(x))$$

$$\int e^x \, sin(x) \, dx = \frac{e^x(sin(x) - cos(x))}{2} + c$$

Integration by parts is a versatile and essential technique in calculus, particularly useful for integrating products of functions where one function can be easily differentiated, and the other can be easily integrated. By mastering the integration by parts formula and practicing with various types of integrands, one can solve a wide range of integral problems effectively. This technique is widely used in applications across mathematics, physics, engineering, and other scientific fields.

3. INTEGRATION OF RATIONAL FUNCTION

Integration of rational functions is a fundamental topic in calculus. Rational functions are ratios of polynomials, and their integration often requires specific techniques such as partial fraction decomposition. This guide provides an overview of the methods and examples related to the integration of rational functions.

3.1. Understanding Rational Functions

A rational function is a function that can be expressed as the ratio of two polynomials:

$$f(x) = \frac{P(x)}{Q(x)}$$

Where P(x) and Q(x) are polynomials.

3.2. Techniques for Integrating Rational Functions

1. **Direct Integration:** If the rational function is simple, it can sometimes be integrated directly.
2. **Long Division:** When the degree of the numerator $P(x)$ is greater than or equal to the degree of the denominator $Q(x)$, polynomial long division can be used to simplify the integrand.
3. **Partial Fraction Decomposition:** For more complex rational functions, partial fraction decomposition is a powerful technique to break down the integrand into simpler fractions that can be integrated individually.

3.3. Direct Integration

In some cases, the rational function is straightforward and can be integrated directly.

Integrate $\int \frac{1}{x} dx$

Solution:

$$\int \frac{1}{x} dx = ln|x| + c$$

3.4. Polynomial Long Division

When the degree of the numerator is greater than or equal to the degree of the denominator, polynomial long division can be used to simplify the rational function.

Example: Integrate $\int \frac{x^2+2x+1}{x+1} dx$

Solution:

Perform Polynomial Long Division:

$$\frac{x^2+2x+1}{x+1} = x+1$$

Integrate the result:

$$\int (x+1)\, dx = \int x\, dx + \int 1\, dx = \frac{x^2}{2} + x + c$$

3.5. Partial Fraction Decomposition

For more complex rational functions, partial fraction decomposition is used to break down the integrand into simpler fractions.

Steps for Partial Fraction Decomposition

1. Factor the denominator: Factor Q(x) into irreducible polynomials.
2. Set up partial fractions: Express $\frac{P(x)}{Q(x)}$ as a sum of simpler fractions.
3. Determine constants: Solve for the constants in the partial fractions.
4. Integrate each term: Integrate each partial fraction individually.

Example: Integrate $\int \frac{2x+3}{(x-1)(x+2)} dx$

Solution:

Set up Partial Fractions:

$$\frac{2x+3}{(x-1)(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x+2)}$$

Determine constants A and B:

$$2x+3 = A(x+2) + B(x-1)$$

$$2x+3 = Ax + 2A + Bx - B$$

$$2x+3 = (A+B)x + (2A-B)$$

Solving for A and B:

$$A + B = 2$$

$$2A - B = 3$$

Adding these equations:

$$3A = 5$$

$$A = \frac{5}{3}$$

Substituting A back:

$$\frac{5}{3} + B = 2$$

$$B = 2 - \frac{5}{3} = \frac{1}{3}$$

Integrate each term:

$$\int \frac{2x+3}{(x-1)(x+2)} dx = \int \frac{5}{3(x-1)} + \frac{1}{3(x+2)}\, dx$$

$$= \frac{5}{3}\int \frac{1}{x-1} dx + \frac{1}{3}\int \frac{1}{x+2} dx$$

$$\frac{5}{3} ln|x-1| + \frac{1}{3} ln|x+2| + c$$

3.6. Integrating Rational Functions with Repeated Linear Factors

When the denominator has repeated linear factors, the partial fraction decomposition needs to account for each repetition.

Example: $\int \frac{4x+5}{(x+1)^2} dx$

Solution:

Set up Partial Fractions:

$$\frac{4x+5}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}$$

Determine Constants A and B:

$$4x + 5 = A(x+1) + B$$

$$4x + 5 = Ax + A + B$$

Solving for A and B:

$$A = 4$$

$$A + B = 5$$

$$4 + B = 5$$

$$B = 1$$

Integrate each term:

$$\int \frac{4x+5}{(x+1)^2}dx = \int \frac{4}{x+1} + \frac{1}{(x+1)^2}dx$$

$$= 4\int \frac{1}{x+1}dx + \int (x+1)^{-2}\,dx$$

$$= 4\,ln|x+1| - \frac{1}{x+1} + c$$

Integration of rational functions often requires specific techniques such as polynomial long division and partial fraction decomposition. These methods break down complex integrals into simpler parts that can be integrated individually. Mastery of these techniques is essential for solving a wide range of problems in calculus and its applications in science, engineering, and other fields. By understanding and practicing these methods, one can effectively tackle integrals involving rational functions.

4. WORKING EXAMPLES

1. Evaluate $I = \int x \cos x \, dx$

   Solution:

   we take $1^{st}\ funciion = f(x) = x$ and $2^{nd}\ funciion = g(x) = \cos x$

   We know integration by parts as

$$\int f(x) \cdot g(x)\, dx = f(x) \cdot \int g(x)\, dx - \left[\int \frac{d}{dx}[f(x)] \cdot \left(\int g(x)\, dx\right) dx\right]$$

$$\Rightarrow \int x \cdot \cos x\, dx = x \cdot \int \cos x\, dx - \left[\int \frac{d}{dx}(x) \cdot \left(\int \cos x\, dx\right) dx\right]$$

$$\Rightarrow \int x \cdot \cos x\, dx = x \cdot \sin x - \left[\int \frac{d}{dx}(x) \cdot (\sin x)dx\right]$$

$$\Rightarrow \int x \cdot \cos x\, dx = x \cdot \sin x - \left[\int 1 \cdot (\sin x)dx\right]$$

$$\Rightarrow \int x \cdot \cos x\, dx = x \cdot \sin x - \left[\int (\sin x)\, dx\right]$$

$$\Rightarrow \int x \cdot \cos x\, dx = x \cdot \sin x - (-\cos x) + C$$

$$\Rightarrow \int x \cdot \cos x\, dx = x \cdot \sin x + \cos x + C$$

2. Evaluate $I = \int \log x$

   Solution:

   We note that the given function is not the product of two functions. To apply ILATE rule we need to have two functions. So, we take

   $1^{st}\ funciion = f(x) = \log x$ and

$$2^{nd}\ funciion = g(x) = 1$$

   We know integration by parts as

$$\int f(x) \cdot g(x)\, dx = f(x) \cdot \int g(x)\, dx - \left[\int \frac{d}{dx}[f(x)] \cdot \left(\int g(x)\, dx\right) dx\right]$$

$$\Rightarrow \int \log x \cdot 1dx = logx \cdot \int 1\, dx \; - \left[\int \frac{d}{dx}(\log x) \cdot \left(\int 1\, dx\right) dx \;\right]$$

$$\Rightarrow \int \log x \cdot 1dx = logx \cdot x \; - \left[\int \frac{1}{x} \cdot xdx \;\right]$$

$$\Rightarrow \int \log x \cdot 1dx = logx \cdot x \; - \left[\int 1\, dx \;\right]$$

$$\Rightarrow \int \log x \cdot 1dx = logx \cdot x \; - x + C$$

3. Evaluate $I = \int x\, tan^{-1}x$

   Solution:

   We note that the given function is the product of two functions namely x an algebraic function and $\tan^{-1} x$ an inverse function. By ILATE rule we take

   $1^{st}\ funciion = f(x) = \tan^{-1} x$ and

$$2^{nd}\ funciion = g(x) = x$$

   We know integration by parts as

$$\int f(x) \cdot g(x)\, dx = f(x) \cdot \int g(x)\, dx \; - \left[\int \frac{d}{dx}[f(x)] \cdot \left(\int g(x)\, dx\right) dx \;\right]$$

$$\Rightarrow \int \tan^{-1} x \cdot xdx = \tan^{-1} x \cdot \int x\, dx \; - \left[\int \frac{d}{dx}(\tan^{-1} x) \cdot \left(\int x\, dx\right) dx \;\right]$$

$$\Rightarrow \int \tan^{-1} x \cdot xdx = \tan^{-1} x \cdot \frac{x^2}{2} \; - \left[\int \frac{1}{1+x^2} \cdot \left(\frac{x^2}{2}\right) dx \;\right]$$

$$\Rightarrow \int \tan^{-1} x \cdot xdx = \tan^{-1} x \cdot \frac{x^2}{2} \; - \frac{1}{2}\left[\int \frac{x^2+1-1}{1+x^2} dx \;\right]$$

$$\Rightarrow \int \tan^{-1} x \cdot xdx = \tan^{-1} x \cdot \frac{x^2}{2} \; - \frac{1}{2}\left[\int \left(\frac{(x^2+1)}{1+x^2} - \frac{1}{1+x^2}\right) dx \;\right]$$

$$\Rightarrow \int \tan^{-1} x \cdot xdx = \tan^{-1} x \cdot \frac{x^2}{2} \; - \frac{1}{2}\left[\int \left(1 - \frac{1}{1+x^2}\right) dx \;\right]$$

$$\Rightarrow \int \tan^{-1} x \cdot xdx = \tan^{-1} x \cdot \frac{x^2}{2} \; - \frac{1}{2}[x - \tan^{-1} x] + C$$

4. Evaluate $I = \int \log x \ \cdot x\, dx$

   Solution:

   We note that the given function is the product of two functions namely x an algebraic function and $\log x$ a logarithmic function. By ILATE rule we take

   $1^{st}\ funciion = f(x) = \log x$ and

$$2^{nd}\ funciion = g(x) = x$$

   We know integration by parts as

$$\int f(x) \cdot g(x)\, dx = f(x) \cdot \int g(x)\, dx \ - \left[\int \frac{d}{dx}[f(x)] \cdot \left(\int g(x)\, dx\right) dx \ \right]$$

$$\Rightarrow \int \log x \cdot x\, dx = \log x \cdot \int x\, dx \ - \left[\int \frac{d}{dx}[\log x] \cdot \left(\int x\, dx\right) dx \ \right]$$

$$\Rightarrow \int \log x \cdot x\, dx = \log x \cdot \frac{x^2}{2} \ - \left[\int \frac{1}{x} \cdot \frac{x^2}{2} dx \ \right]$$

$$\Rightarrow \int \log x \cdot x\, dx = \log x \cdot \frac{x^2}{2} \ - \frac{1}{2}\left[\int x\, dx \ \right]$$

$$\Rightarrow \int \log x \cdot x\, dx = \log x \cdot \frac{x^2}{2} \ - \frac{1}{2}\left[\frac{x^2}{2}\right] + C$$

$$\Rightarrow \int \log x \cdot x\, dx = \log x \cdot \frac{x^2}{2} \ - \frac{x^2}{4} + C$$

5. Evaluate: $I = \int x^2 \ \cdot \log x \ dx$

   Solution:

   We note that the given function is the product of two functions namely x^2 an algebraic function and $\log x$ a logarithmic function. By ILATE rule we take

   $1^{st}\ funciion = f(x) = \log x$ and

$$2^{nd}\ funciion = g(x) = x^2$$

   We know integration by parts as

$$\int f(x)\cdot g(x)\,dx = f(x)\cdot\int g(x)\,dx - \left[\int \frac{d}{dx}[f(x)]\cdot\left(\int g(x)\,dx\right)dx\right]$$

$$\Rightarrow \int \log x\cdot x^2\,dx = \log x\cdot\int x^2\,dx - \left[\int \frac{d}{dx}[\log x]\cdot\left(\int x^2\,dx\right)dx\right]$$

$$\Rightarrow \int \log x\cdot x^2\,dx = \log x\cdot\frac{x^3}{3} - \left[\int \frac{1}{x}\cdot\frac{x^3}{3}dx\right]$$

$$\Rightarrow \int \log x\cdot x^2\,dx = \log x\cdot\frac{x^3}{3} - \frac{1}{3}\left[\int x^2 dx\right]$$

$$\Rightarrow \int \log x\cdot x^2\,dx = \log x\cdot\frac{x^3}{3} - \frac{1}{3}\left(\frac{x^3}{3}\right) + C$$

$$\Rightarrow \int \log x\cdot x^2\,dx = \log x\cdot\frac{x^3}{3} - \frac{x^3}{9} + C$$

6. Evaluate: $I = \int x \cdot \sec^2 x\ dx$

   Solution:

   We note that the given function is the product of two functions namely x an algebraic function and $\sec^2 x$ a trigonometric function. By ILATE rule we take

   $1^{st}\ funciion = f(x) = x$ and

$$2^{nd}\ funciion = g(x) = \sec^2 x$$

   We know integration by parts as

$$\int f(x)\cdot g(x)\,dx = f(x)\cdot\int g(x)\,dx - \left[\int \frac{d}{dx}[f(x)]\cdot\left(\int g(x)\,dx\right)dx\right]$$

$$\int x\cdot\sec^2 x\ dx = x\cdot\int \sec^2 x\,dx - \left[\int \frac{d}{dx}[x]\cdot\left(\int \sec^2 x\,dx\right)dx\right]$$

$$\Rightarrow \int x\cdot\sec^2 x\ dx = x\cdot\tan x - \left[\int 1\cdot\tan x\,dx\right]$$

$$\Rightarrow \int x\cdot\sec^2 x\ dx = x\cdot\tan x - \left[\int \tan x\,dx\right]$$

$$\Rightarrow \int x\cdot\sec^2 x\ dx = x\cdot\tan x - \log(\sec x) + C$$

7. Evaluate: $I = \int x \cdot \operatorname{cosec}^2 x \; dx$

Solution:

We note that the given function is the product of two functions namely x an algebraic function and $\operatorname{cosec}^2 x$ a trigonometric function. By ILATE rule we take

$1^{st}\ funciion = f(x) = x$ and

$$2^{nd}\ funciion = g(x) = \operatorname{cosec}^2 x$$

We know integration by parts as

$$\int f(x) \cdot g(x)\, dx = f(x) \cdot \int g(x)\, dx - \left[\int \frac{d}{dx}[f(x)] \cdot \left(\int g(x)\, dx\right) dx\right]$$

$$\Rightarrow \int x \cdot \operatorname{cosec}^2 x \; dx = x \cdot \int \operatorname{cosec}^2 x\, dx - \left[\int \frac{d}{dx}[x] \cdot \left(\int \operatorname{cosec}^2 x\, dx\right) dx\right]$$

$$\Rightarrow \int x \cdot \operatorname{cosec}^2 x \; dx = x \cdot (-\cot x) - \left[\int 1 \cdot (-\cot x)\, dx\right]$$

$$\Rightarrow \int x \cdot cosec^2 x \; dx = x \cdot (-\cot x) - \left[\int (-\cot x)\, dx\right]$$

$$\Rightarrow \int x \cdot \operatorname{cosec}^2 x \; dx = -x \cot x + \log(\sin x) + C$$

8. Evaluate: $I = \int e^x \cdot \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$

Solution:

We note that the given integral contains product of two functions, first one being exponential function. We know

$$\int e^x \big(f(x) + f'(x)\big) dx = e^x\big(f(x)\big) + C$$

In this case, we take $f(x) = \frac{1}{x}$ and note that $f'(x) = -\frac{1}{x}^2$. Hence by applying formula, we get,

$$I = \int e^x \cdot \left(\frac{1}{x} - \frac{1}{x^2}\right) dx = e^x \left(\frac{1}{x}\right) + C$$

9. Evaluate: $I = \int e^x \cdot (\sin x + \cos x)\, dx$

Solution:

We note that the given integral contains product of two functions, first one being exponential function. We know

$$\int e^x\big(f(x) + f'(x)\big)dx = e^x\big(f(x)\big) + C$$

In this case, we take $f(x) = \sin x$ and note that $f'(x) = \cos x$. Hence by applying formula, we get,

$$I = \int e^x \cdot (\sin x + \cos x)\, dx = e^x \sin x + C$$

10. Evaluate: $I = \int e^x \cdot \left(\frac{1+\sin x}{1+\cos x}\right) dx$

Solution:

We note that the given integral contains product of two functions, first one being exponential function. We know

$$\int e^x\big(f(x) + f'(x)\big)dx = e^x\big(f(x)\big) + C$$

Consider

$$\frac{1+\sin x}{1+cosx} = \frac{1}{1+\cos x} + \frac{\sin x}{1+\cos x} = \left(\frac{1}{2\cos^2\left(\frac{x}{2}\right)} + \frac{2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}{2\cos^2\left(\frac{x}{2}\right)}\right)$$

$$= \left(\frac{1}{2}\sec^2\left(\frac{x}{2}\right) + \tan\left(\frac{x}{2}\right)\right)$$

Thus,

$$I = \int e^x \cdot \left(\frac{1+\sin x}{1+\cos x}\right) dx = \int e^x \cdot \left(\frac{1}{2}\sec^2\left(\frac{x}{2}\right) + \tan\left(\frac{x}{2}\right)\right) dx$$

$$\Rightarrow \quad I = \int e^x \cdot \left(\tan\left(\frac{x}{2}\right) + \frac{1}{2}\sec^2\left(\frac{x}{2}\right)\right) dx$$

If we take

$$f(x) = \tan\left(\frac{x}{2}\right)\ ,\ then\ f'(x) = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)$$

Thus,

$$I = \int e^x \cdot \left(\frac{1+\sin x}{1+\cos x}\right) dx = \int e^x \cdot \left(\tan\left(\frac{x}{2}\right) + \frac{1}{2}\sec^2\left(\frac{x}{2}\right)\right) dx = e^x \tan\left(\frac{x}{2}\right) + C$$

11. Evaluate: $\int \frac{x}{(x+1)(x+2)} dx$

Solution:

The function within the integral is $\frac{x}{(x+1)(x+2)}$. The denominator is the product of two non-repeated linear factors. We apply partial fractions as below:

$$\frac{x}{(x+1)(x+2)} = \frac{A}{(x+1)} + \frac{B}{(x+2)}$$

Multiplying both sides by $(x+1)(x+2)$ we get,

$$x = A(x+2) + B(x+1) ----------< 1 >$$

Put $x = -1\ in < 1 >$, $\quad -1 = A(-1+2) + B(-1+1)$

$$\Rightarrow -1 = A \quad \Rightarrow A = -1$$

Put $x = -2\ in < 1 >$, $\quad -2 = A(-2+2) + B(-2+1)$

$$\Rightarrow -2 = -B \quad \Rightarrow B = 2$$

Thus, $\frac{x}{(x+1)(x+2)} = \frac{-1}{(x+1)} + \frac{2}{(x+2)}$

$$\Rightarrow \int \frac{x}{(x+1)(x+2)} dx = \int \frac{-1}{(x+1)} dx + \int \frac{2}{(x+2)} dx$$

$$\Rightarrow \int \frac{x}{(x+1)(x+2)} dx = -\log(x+1) + 2\log(x+2) + C$$

$$\Rightarrow \int \frac{x}{(x+1)(x+2)} dx = -\log(x+1) + \log(x+2)^2 + C$$

$$\Rightarrow \int \frac{x}{(x+1)(x+2)} dx = \log\left(\frac{(x+2)^2}{x+1}\right) + C$$

12. Evaluate: $\int \frac{1}{x^2-9} dx$

Solution:

The function within the integral is $\frac{1}{x^2-9} = \frac{1}{(x-3)(x+3)}$. The denominator is the product of two non-repeated linear factors. We apply partial fractions as below:

$$\frac{1}{(x-3)(x+3)} = \frac{A}{(x-3)} + \frac{B}{(x+3)}$$

Multiplying both sides by $(x-3)(x+3)$ we get,

$$1 = A(x+3) + B(x-3) ----------< 1 >$$

Put $x = 3\ in < 1 >$, $\quad 1 = A(3+3) + B(3-3)$

$$\Rightarrow 1 = 6A \quad \Rightarrow A = \frac{1}{6}$$

Put $x = 3\ in < 1 >$, $\quad 1 = A(3-3) + B(-3-3)$

$$\Rightarrow 1 = -6B \quad \Rightarrow B = -\frac{1}{6}$$

Thus,

$$\frac{1}{(x-3)(x+3)} = \frac{\frac{1}{6}}{(x-3)} + \frac{\frac{-1}{6}}{(x+3)}$$

$$\Rightarrow \int \frac{1}{(x-3)(x+3)} dx = \frac{1}{6}\int \frac{1}{x-3} dx - \frac{1}{6}\int \frac{1}{x+3} dx$$

$$\Rightarrow \int \frac{1}{(x-3)(x+3)} dx = \frac{1}{6}\log(x-3) - \frac{1}{6}\log(x+3) + C$$

$$\Rightarrow \int \frac{1}{(x-3)(x+3)} dx = \frac{1}{6}(\log(x-3) - \log(x+3)) + C$$

$$\Rightarrow \int \frac{1}{(x-3)(x+3)} dx = \frac{1}{6}\log\left(\frac{x-3}{x+3}\right) + C$$

13. Evaluate: $\int \frac{2x+3}{x^2-5x+6} dx$

Solution:

The function within the integral is $\frac{2x+3}{x^2-5x+6} = \frac{2x+3}{(x-3)(x-2)}$. The denominator is the product of two non-repeated linear factors. We apply partial fractions as below:

$$\frac{2x+3}{(x-3)(x-2)} = \frac{A}{(x-3)} + \frac{B}{(x-2)}$$

Multiplying both sides by $(x-3)(x-2)$ we get,

$$2x+3 = A(x-2) + B(x-3) - - - - - - - - - < 1 >$$

Put $x = 3\ in < 1 >$, $\quad 9 = A(3-2) + B(3-3)$

$$\Longrightarrow 9 = A$$

Put $x = 2\ in < 1 >$, $\quad 7 = A(2-2) + B(2-3)$

$$\Longrightarrow 7 = -B \qquad \Longrightarrow B = -7$$

Thus,

$$\frac{2x+3}{(x-3)(x-2)} = \frac{9}{(x-3)} + \frac{-7}{(x-2)}$$

$$\Longrightarrow \int \frac{2x+3}{(x-3)(x-2)} dx = \int \frac{9}{x-3} dx + \int \frac{-7}{x+3} dx$$

$$\Longrightarrow \int \frac{2x+3}{(x-3)(x-2)} dx = 9 \int \frac{1}{x-3} dx - 7 \int \frac{-1}{x+3} dx$$

$$\Longrightarrow \int \frac{2x+3}{(x-3)(x-2)} dx = 9 \log(x-3) - 7 \log(x+3) + C$$

14. Evaluate: $\int \frac{3x+5}{x^3-x^2-x+1} dx$

Solution:

Consider

$$x^3 - x^2 - x + 1 = x^2\,(x-1) - 1(x-1) = (x-1)(x^2-1)$$

$$= (x-1)(x-1)(x+1) = (x-1)^2(x+1)$$

The function within the integral is $\frac{3x+5}{x^3-x^2-x+1} = \frac{3x+5}{(x-1)^2(x+1)}$. The denominator is the product of one repeated and one non-repeated linear factor. We apply partial fractions as below:

$$\frac{3x+5}{(x-1)^2(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+1)}$$

Multiplying both sides by $(x-1)^2(x+1)$ we get,

$$3x+5 = A(x-1)(x+1) + B(x+1) + C(x-1)^2 ---------< 1 >$$

Put $x = 1\ in < 1 >$, $\quad 8 = B(2) \quad \Rightarrow B = 4$

Put $x = -1\ in < 1 >$, $\quad 2 = C(4) \quad \Rightarrow C = \frac{1}{2}$

Equating co-efficient of x^2, we get, $A + B = 0 \quad \Rightarrow A = -B = -4$

$$\frac{3x+5}{(x-1)^2(x+1)} = \frac{-4}{(x-1)} + \frac{4}{(x-1)^2} + \frac{\frac{1}{2}}{(x+1)}$$

$$\Rightarrow \int \frac{3x+5}{(x-1)^2(x+1)}dx = \int \frac{-4}{(x-1)}dx + \int \frac{4}{(x-1)^2}dx + \frac{1}{2}\int \frac{1}{(x+1)}dx$$

$$\Rightarrow \int \frac{3x+5}{(x-1)^2(x+1)}dx = -4\log(x-1) - \frac{4}{x-1} + \frac{1}{2}\log(x+1) + C$$

15. Evaluate: $\int \frac{x}{(x-1)(x^2+1)}dx$

Solution:

The function within the integration is $\frac{x}{(x-1)(x^2+1)}$.The denominator is the product of $(x-1)$ a linear factor and (x^2+1) is a quadratic factor. We apply partial fractions as below:

$$\frac{x}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

Multiplying both sides by $(x-1)(x^2+1)$ we get,

$$x = A(x^2+1) + (Bx+C)(x-1) ----------< 1 >$$

Prut $x = 1 \ in \ < 1 >$ we get, $1 = 2A \quad \Longrightarrow A = \frac{1}{2}$

Also, $x = Ax^2 + A + Bx^2 - Bx + Cx - C$

Equating co-efficient of x^2 we get, $0 = A + B \quad \Longrightarrow 0 = \frac{1}{2} + B \quad \Longrightarrow B = -\frac{1}{2}$

Equating co-efficient of x we get, $1 = C - B \quad \Longrightarrow 1 = C + \frac{1}{2} \quad \Longrightarrow C = \frac{1}{2}$

Thus,

$$\frac{x}{(x-1)(x^2+1)} = \frac{\frac{1}{2}}{x-1} + \frac{-\frac{1}{2}x + \frac{1}{2}}{x^2+1}$$

$$\Longrightarrow \int \frac{x}{(x-1)(x^2+1)} dx = \int \frac{\frac{1}{2}}{x-1} + \int \frac{-\frac{1}{2}x + \frac{1}{2}}{x^2+1} dx$$

$$\Longrightarrow \int \frac{x}{(x-1)(x^2+1)} dx = \frac{1}{2}\int \frac{1}{x-1} - \frac{1}{2}\int \frac{x}{x^2+1} dx + \frac{1}{2}\int \frac{1}{x^2+1} dx$$

$$\Longrightarrow \int \frac{x}{(x-1)(x^2+1)} dx = \frac{1}{2}\log(x-1) - \frac{1}{4}\log(x^2+1) + \frac{1}{2}\tan^{-1}(x) + K$$

5. SUMMARY

After a thorough journey through Unit 13 on Integration Part II, we've deepened our understanding of integration techniques, focusing primarily on Integration by Parts—a powerful method that extends beyond basic integration skills. This technique is not just a mathematical routine; it's a strategic tool that enhances our ability to handle functions that are products of simpler elements, which might otherwise seem daunting.

Integration by Parts is essentially derived from the product rule for differentiation and is particularly useful when the integrand is a product of two functions. One of these functions is usually easy to differentiate, while the other is easier to integrate. Remembering the formula, $\int u dv = uv - \int v du$, is crucial, but understanding how to select u and dv effectively can significantly simplify your integration process.

We've tackled various examples in this unit to solidify your grasp of this technique. For instance, integrating expressions like xe^x or x lnf_0(x) demonstrates the practical utility of Integration by Parts. By choosing u and dv strategically, we transform these complex integrals into simpler forms that we can manage more straightforwardly.

Moreover, this unit isn't just about following steps mechanically. It's about developing a keen eye for the structure of mathematical expressions and leveraging the properties of derivatives and integrals to solve problems efficiently. As we applied this method to different functions, including trigonometric and exponential functions, we saw how versatile and effective Integration by Parts could be.

In practical terms, whether it's in physics, engineering, or economics, the ability to integrate complex functions accurately and efficiently is invaluable. This skill set allows you to calculate anything from the work done in a physical system to the accumulated growth in a financial model.

As we wrap up this unit, consider the variety of exercises we've gone through as building blocks. Each example has not only helped to reinforce the method and its steps but also encouraged you to think critically about how and when to apply this technique. Looking forward, continue practicing with diverse functions to become more confident and intuitive

about choosing u and dv in Integration by Parts, and remember—practice makes perfect! Integration is a vast field, and mastery comes with patience and persistent effort.

INSPIRED BY LIFE

6. SELF-ASSESSMENT QUESTIONS

1. The formula for integration by Parts is $\int u\,dv =$ _____ $-\int v\,dv$
2. In the integration by parts formula, if u = x and $dv = e^x\,dx$, then v equals _____.
3. When integrating $x\,sin(x)$ by parts, one should choose u = _____ and dv = _____.
4. If you integrate x^2 by differentiating it in the integration by parts process, du would be _____.
5. The integral $\int\ ln(x)\,dx$ by parts results in _____ . x - $\int$ ____ dx.
6. What is v if $dv = cos(x)\,dx$?
   a. $sin(x)$
   b. $-\,sin(x)$
   c. $\tan(x)$
   d. $-\,tan(x)$
7. Which function should ideally be chosen as u when using integration by parts for $\int xe^{x^2}\,dx$?
   a. x
   b. e^{x^2}
   c. x^2
   d. $2x$
8. What is the result of integrating $\int e^x\,sin(x)\,dx$ using integration by parts twice?
   a. $\frac{1}{2}e^x(sin(x) - cos(x)) + c$
   b. $e^x(\sin(x) + \cos(x)) + c$
   c. $e^x \sin(x) - e^x \cos(x) + c$
   d. $e^x \sin(x) + c$
9. *the expression* $\int x^3 \ln(x)\,dx$ is best solved by setting u = _____.
   a. x^3
   b. $\ln(x)$
   c. $3x^2$
   d. $\frac{1}{x}$
10. When applying Integration by parts to $\int\ ln(x) \cdot x\,dx$, which if the following integrals appears in the solution?

a. $\int x\,dx$

b. $\int \frac{1}{x}dx$

c. $\int \frac{x\,dx}{\ln(x)}$

d. $\int x\ln(x)\,dx$

7. TERMINAL QUESTIONS

1. Compute $\int xe^x dx$ using integration by parts.
2. Solve $\int x^2 \ln(x) dx$ by applying integration by parts.
3. Evaluate $\int \cos(x) e^x dx$ using the method of integration by parts.
4. Find $\int x\sin(x) dx$ by applying integration by parts.
5. Determine $\int \ln(x) dx$ using integration by parts.
6. Calculate $\int x^3 e^{-x} dx$ by using integration by parts.
7. Solve $\int \tan^{-1}(x) dx$ using integration by parts.
8. Compute $\int x^2 e^{2x} dx$ through integration by parts.
9. Evaluate $\int \sin(x)\cos(x) dx$ using integration by parts.
10. Determine $\int xe^{x^2} dx$ by applying a suitable substitution followed by integration by parts if necessary.
11. Calculate $\int x\ln(x^2) dx$ using integration by parts.
12. Find $\int \frac{x}{e^x} dx$ using integration by parts.
13. Solve $\int x^2 \sin(x) dx$ through integration by parts.
14. Compute $\int e^x \sin(x) dx$ by using integration by parts twice.
15. Evaluate $\int \log_{10}(x) dx$ using integration by parts.
16. Determine $\int x\cos^2(x) dx$ using integration by parts.
17. Calculate $\int xe^{3x} dx$ using integration by parts.
18. Find $\int \sqrt{x} e^x dx$ through integration by parts.
19. Solve $\int x^2 \cos(2x) dx$ using integration by parts.
20. Compute $\int \frac{\ln(x)}{x^2} dx$ by applying integration by parts.

8. ANSWERS

Self-Assessment Questions

1. $uv - \int v\, du$
2. e^x
3. $x, \sin(x)\, dx$
4. $2x$
5. $x\, ln(x)\, , x$
6. B. $-\, sin(x)$
7. A. x
8. A. $\frac{1}{2}e^x(sin(x) - cos(x)) + c$
9. B. $ln(x)$
10. B. $\int \frac{1}{x} dx$

Terminal Questions

1. $xe^x - e^x + C$
2. $\frac{x^2 \ln(x)}{2} - \frac{x^2}{4} + C$
3. $e^x \cos(x) + e^x \sin(x) + C$
4. $x\sin(x) + \cos(x) + C$
5. $x\ln(x) - x + C$
6. $-x^3 e^{-x} - 3x^2 e^{-x} - 6xe^{-x} - 6e^{-x} + C$
7. $x\tan^{-1}(x) - \frac{1}{2}\ln(1 + x^2) + C$
8. $e^{2x}\left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{4}\right) + C$
9. $\frac{1}{2}\sin^2(x) + C$
10. Using substitution $u = x^2, \frac{1}{2}e^{x^2} + C$
11. $x^2 \ln(x^2) - x^2 + C$
12. $-\frac{x}{e^x} - \frac{1}{e^x} + C$

13. $-x^2\cos(x) + 2x\sin(x) + 2\cos(x) + C$

14. $\frac{e^x}{2}(\sin(x) - \cos(x)) + C$

15. $x\log_{10}(x) - (\ln(10)x) + C$

16. $x\frac{\sin(2x)}{4} + \frac{x^2}{2} + C$

17. $\frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x} + C$

18. $2\sqrt{x}e^x - 2e^x\sqrt{x} + C$

19. $\frac{x^2\sin(2x)}{2} + \frac{x\cos(2x)}{2} - \frac{\sin(2x)}{4} + C$

20. $-\frac{\ln(x)}{x} - \frac{1}{x} + C$